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A Comparative Survey on Parts of Speech Taggers for the Marathi Language

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***Abstract*—Natural Language Processing relies heavily on the POS tagger. The POS tagger is a useful tool for tagging each word in a phrase with parts of speech tags. NLP Applications performing various tasks use POS tagging as a crucial initial step. In terms of data tagging, the POS tagger for English is widely available, however, there is no similar tagger for Marathi. Marathi is a morphologically complex language with regional speech differences. Because of the ambiguity in the language, as well as its highly inflectional structure and free word order, establishing a successful POS tagger in Marathi is tough. This article provides a comprehensive overview of POS tagging for the Marathi language and its variations. Various POS Tagging models and approaches are investigated in this research. Tokenization in computer languages is similar to tagging in natural language processing. Choosing the appropriate tag for the scenario might be challenging for POS taggers. Research has been conducted to discover a solution to this conundrum.**

***Index Terms*—Natural Language Processing, Marathi, Part-Of-Speech Tagging, Hidden Markov Model, N-grams.**

I. Introduction

The POS tagger facilitates the assignment of tags to the words in a phrase. The POS system provides a ton of information about the words. Syntax, semantics, and translation are just a few of the NLP tasks that are taken into account when processing a language. Corpus linguistics and POS tagging research are inextricably linked.

According to statistics, Marathi is spoken by a significant number of people in India. In recent years many research institutions and organizations are involved in developing the lexical resources for Marathi for NLP activities. Assigning the right tag to a given word is defined as POS tagging. POS tagging is one of the toughest challenge when attempted in rich languages like Marathi. There is a large variety of text data accessible in Marathi, thus it is necessary to sort it properly. It is a crucial aspect of interpreting natural language in linguistics. The ambiguity of the Marathi language makes it difficult to accurately tag the term.

Unsupervised and supervised tagging approaches are the two types of tagging techniques [1]. A previously trained corpus is incorporated into the supervised technique, and the output is created based on it. It is simple to tag a sentence using the supervised method. The unsupervised technique's first goal is to train the whole corpus, following which the input is analyzed. This approach is more difficult to use than the supervised tagging method. The three primary kinds are stochastic or statistical, rule-based, and hybrid-based techniques. A regularly used previous approach was the rule-based method [2], which uses information and rules created manually. The researchers struggled to come up with appropriate rules. A rule-based strategy does not always function effectively. One of the major flaws of the rule-based strategy is that it fails to generate rules for words that are unknown. The accurate tag will not be considered in a rule-based approach. An extensive set of handwritten rules is employed to achieve good performance for this system.

TABLE I
POS Tags

SR. NO.	TAGS	DESCRIPTION
1	<NN>	Noun
2	<NNC>	Compound Noun
3	<NNP>	Proper Noun
4	<PRP>	Pronoun
5	<ADJ>	Adjective
6	<ADV>	Adverb
7	<CNJ>	Conjunction
8	<VN>	Verbal Noun
9	<Q>	Quantifier
10	<DET>	Determiners
11	<PSP>	Post Positions
12	<RP>	Particles
13	<RD_PUNC>	Punctuations

Using a statistical approach, researchers produced improved POS taggers. Standardization is one of the primary challenges; different researchers have utilised different tag sets. Statistical approaches, it is made up of frequency and probability. The statistical technique uses unlabelled training data to identify the commonly used tag for a given word, which is then used to assign the tag to the word. It creates a string of tags for the input that isn't acceptable as per the language's grammatical rules. The probabilistic properties of the statistical technique are primarily used in the hybrid method [3], which is then

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followed by the application of a set of handwritten rules. Hybrid performs better than rule-based and statistical models.

HMM is a sequence labeling algorithm that calculates the probability distribution across all potential label sequences and selects the optimal label sequence from a set of units [4]. HMM discusses input words as observed events and pos tags as hidden events. CRF is a type of undirected graphical model [5]. The fundamental advantage of CRFs over Hidden Markov models is that they are conditional, allowing the independence requirements required by HMMs to be relaxed, resulting in tractable inference. Ambiguity is a significant problem that must be handled during the development of the tagger. The numerous POS tagging techniques are discussed in this study. The POS is made up of nouns, adjectives, verbs, determiners, adverbs, and other grammatical components. Various tags are shown in Table I

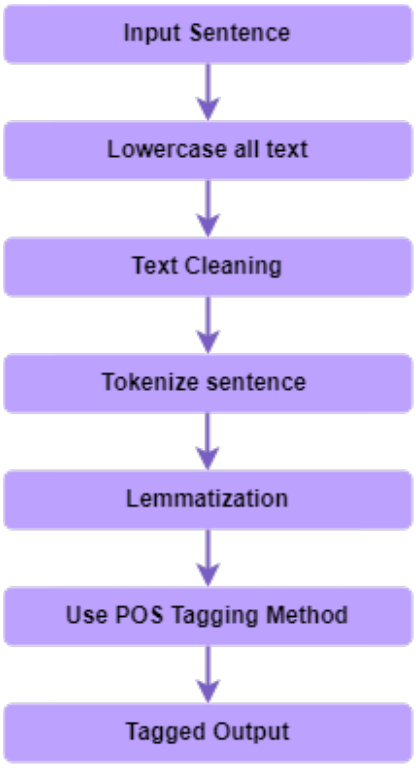

Fig. 1. Part-of-Speech Flowchart

POS tagging is commonly known as grammatical tagging of text in terms of idea and context. For the English language, there are POS taggers such as Tree tagger, Brill tagger, and a few more. Rule-based, stochastic, and morphological inputs were used by these taggers. POS is primarily divided into two categories. The POS tagger is process that scans words and allocates tags to each one. The POS tagging procedure is demonstrated in [Fig. 1] Fig. 1. The input is first converted into the lower case for uniformity and then cleaned to remove any redundancy that can affect the outcome, all of which are done before doing POS tagging in order to produce better results. The cleaned input is subjected to further tokenization. Tokenization refers to splitting sequence of words into individual tokens. The simplest way to perform tokenization is to use standard split functions in any programming language but that may not work for punctuations. To perform more accurate tokenization, python libraries such as NLTK, sklearn and Gensim can be used. This tokenized data is provided to the lemmatizer as input, where the word is changed to its lemma base form while taking into account its dictionary meaning and sense. Each token is then given part of the speech tag. This method is used to apply tags to these tokens. After that, the tagged output is generated. Due to the presence of ambiguity in the Marathi Language, there are certain challenges in POS Tagging, [Fig. 2]Fig. 2 demonstrates such ambiguity with an example.

"राम /NN आणि/CC श्याम /NN एकदा/RB फिरायला /VM गेले/NNP होते/NNP"

The above example shows the incorrect tag for the word गेले/NNP and होते/NNP. The correct tag for that word is the "गेले" as VM and the word "होते" is the VAUX.

Fig. 2. Ambiguity Example

II. Methods

The supervised and unsupervised POS tagging models are discussed in this section. The three major approaches are discussed.

A. Supervised

Frequency or probability are the fundamentals used by statistical taggers to tag the content. Word ambiguity is handled with a basic statistical tagger, which resolves the chance of the word occurring with a specific tag. The training set is made up of these tags, which are assigned to inaccurate instances of the terms when analysing the data. Data linked to the tag set is located using them, pre-tagged models are necessary for supervised tagging models. The dominant tag for a particular word has a likelihood of appearing via n-1 preceding tags, according to N-gram methods.

B. Unsupervised

Unsupervised POS does not necessarily require previously tagged corpora. This method uses the Baum-Welch algorithm which is a computational technique. The assessment of probabilistic information or the development of rules required by rule-based systems or transformation-based systems is carried out by stochastic taggers.

C. Rule Based

This is one of the first word-tagging approaches, and it relies on grammatical understanding and handwritten rules. They are known as context frame rules. Brill's tagger is a popular rule-based English POS Tagger approach. This tagger searches a lexicon or dictionary for specified tags for a word. When there are several alternative tags for a word, handwritten rules are used to decide which one is valid. To resolve the word's ambiguity, the grammatical properties of the phrase,

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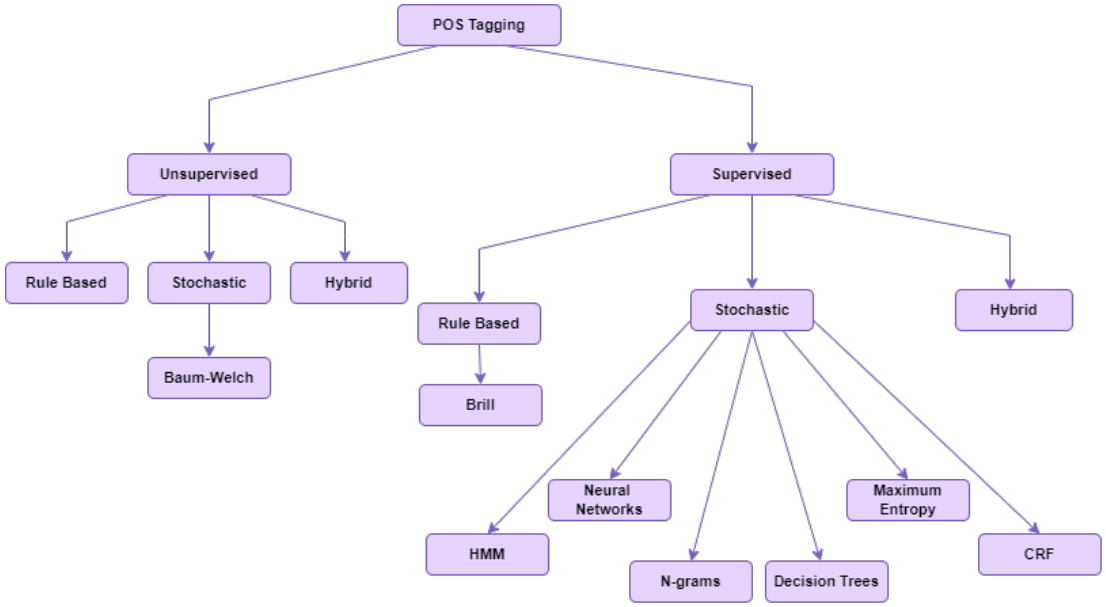

Fig. 3. Part-of-Speech Taggers

its previous word, its following word, and other characteristics are utilised.

D. Stochastic

Probability, statistics, and frequency are all included in a stochastic model. The approaches used in these models include Hidden Markov Models, Maximum Likelihood Estimation, and n-grams. The likelihood of a certain sequence of tags is calculated using the N-gram technique, which can be used instead of the frequency approach. Unigram, bigram, and trigram models are the three types of models.

E. Hybrid

This model assigns the best relevant tag to the phrase using statistical approaches. If it is not correct, the tagger tries to correct it with other rules. It is a combination of rule-based and stochastic POS taggers.

F. N-grams

N-gram is a contiguous sequence of words. Based on the value of N, N-grams can be Unigram, Bigram, or Trigram. Maximum Likelihood Estimation is used to estimate N-grams. N-grams are based on Markov assumption which states that the likelihood of a POS tag is determined by the previous n-1 tags. That is why N-gram is also known as the N-1 order Markov model. e.g. Bigram model is also called as First order Markov model. 1-gram tagger is another name for Unigram tagger. It is based on a straightforward statistical method. It tags each word with the tag that most closely matches the individual term.

$$P(t_i/w_i) = freq(w_i/t_i)/freq(w_i) \tag{1}$$

The word's tag is assigned by the Bigram tagger depending on the previous tag. The Bigram tagger assumes that the tag's likelihood is determined by the tag before it. This method is represented by the equation below.

$$P(t_i/w_i) = P(w_i/t_i).P(t_i/t_{i-1}) \tag{2}$$

Given the current tag, the probability of the current word is P (w_i / t_i). The probability of the current tag is P (t_i / t_{i-1}). The following equation is used to compute these probabilities:

$$P(t_i/t_{i-1}) = f(t_{i-1}, t_i)/f(t_{i-1}) \tag{3}$$

where, w_i is i^{th} word in a sequence and t_i is pos tag of ith word in a sequence

III. Techniques

The following categories can be used to classify supervised and unsupervised tagging techniques.

A. Decision Tree

It is a predictive model that recursively splits the training data set using a tree topology. Each branch of a decision tree reflects an outcome of a test on a feature value. Once a leaf node (or terminal) is reached, a prediction is produced.

B. Hidden Markov Model

State transitions are not visible in Hidden Markov Models. To train an HMM tagger, all that is needed is a vocabulary and some untagged text. HMM is designed to generate a language model with minimal effort. Giving a more likely tag accomplishes disambiguation. A HMM is built up of states corresponding to tags, an alphabet consisting of a set of words, and transition probabilities P[(Tag_i/Tag_{i-1})], and emission probabilities [P($Word_i$/Tag_i)]. In HMM, we have the following probability for a given (word, tag) pair:

$$P(w,t) = \sqcap P(Tag_i|Tag_{i-1}) * P(Word_i|Tag_i) \tag{4}$$

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TABLE II
LITERATURE SURVEY

Study	Methodology	Performance Measures	Features	Weakness
Deshmukh and Kiwelekar [6]	A deep learning approach featuring bidirectional long short-term memory was proposed	Accuracy- HMM - 78, NN-84 CRFs and DL-85 Bi-LSTM (highest)-97 Precision- 0.85 Recall-0.85 F1-0.84 Support-3932	BI-LSTM and deep learning models are compared using various machine learning algorithms.	1500 sentences totaling 10,115 words are used in the experiments, which are short for modelling mentioned approaches. Additionally, the discussed models are not evaluated against cutting-edge research in the same area of study
H B Patil, A.S. Patil, and B.V. Pawar [7]	POS tagging system architecture using the rule-based technique is proposed.	Accuracy - 78.82 Precision - 0.8644 Recall - 0.9272 F-M - 0.8946	Three datasets were used to evaluate the system, with an average accuracy of 78.82%, which is acceptable.	The size of the corpus is comparatively small
Jyoti Singh, Nisheeth Joshi, and Iti Mathur [8]	Tagset developed by IIIT Hyderabad is used. The trigram model is used for Marathi POS tagging.	Accuracy - 91.63%	For automated tagging, the POS tagger described here is quite simple and effective.	It's a bit difficult because of the morphological intricacy of the Marathi Language. In the training corpus, more tagged sentences are required.
Diksha N.Prabhu Khorjuvenkar, Megha Ainapurkar, and Sufola Chagas [9]	The goal of the study was to create a pos tagging method for Konkani considering the Konkani corpus. The finest tag was discovered using the HMM and the Viterbi Algorithm.	Not provided	Since Konkani is a morphologically rich language, there are some issues with the tagging technique, but they are addressed correctly. This labelled data may then be employed in tasks such as information retrieval, speech recognition, and so on.	By employing a new training technique and adding the training data from multiple domains, this POS model for tagging may be enhanced even further.
Rushali Deshmukh [10]	A corpus of 1500 sentences is compiled from Marathi news articles and then manually labelled. The algorithms Naive Bayes, Decision Trees, HMMs, and others are utilised and compared.	Accuracy - 86% Recall - 0.86 Precision - 0.86 F1-score - 0.86 Support - 3840	The key contribution is the creation of a Marathi POS tagged dataset, as well as the assessment of its performance using various models.	The performance can be improved by expanding the labelled dataset, which includes data from several fields. A variety of machine learning algorithms can be used.
Gajanan Rane, Nilesh Joshi, Geetanjali Rane, Hanumant Redkar, Malhar Kulkarni and Pushpak Bhattacharyya [11]	The BIS tag set for Marathi has been discussed. With examples, lexical and functional tagging techniques were explained.	Not provided	The survey in this study is useful for Marathi and other language annotators, scholars, language learners, and so on.	It is necessary to consider issues such as tagging for words with multiple meanings and numerous functional tags. The statistical analysis will be used to evaluate lexical and functional tagging.
Vijayshri Khedkar, Pritesh Shah [12]	The research focuses on POS tagging in the Marathi dialect utilising rule-based tagging, unigram, bigram, and more.	Accuracy - 93%	For Marathi POS tagging, the Hidden Markov model combined with the Viterbi algorithm produced the best results in this paper.	To enhance results, the amount of tagged corpus must be increased, and POS taggers must be trained to use the tagged corpus.
Madhuri M. Deshpande,Dr Sharad D. Gore [3]	This study proposes a hybrid, multi-pass tagger for Marathi sentences, it generates a feature vector for every word by referencing the word before and following the current word being tagged.	Accuracy - 84.63%	Data set created using this tagging approach is cost-effective with respect to space and temporal complexity. The stem may be obtained using morpheme sequencing rules.	To expand the scope of this study by integrating POS tag-derived Marathi phrases. Prefix or affix elimination may also be considered.

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Study	Methodology	Performance Measures	Features	Weakness
Madhav Kankhar, Namrata Mahender, and Kalpana Khandale [13]	Marathi POS tagging is presented in this study. This is a domain-specific document that is based on children's stories.	Accuracy - 67%	When compared to shallow POS taggers, the suggested POS tagger produces better results.	Because it is domain-specific, might be less effective for general Marathi sentences.
Nita V. Patil [14]	To label Marathi text with POS tags, a supervised learning method based on the HMM is used. The dataset used to train the algorithm is made up of 12,000 Marathi sentences taken from a popular Marathi newspaper.	Accuracy - 86.61%.	On test data, the proposed system based on the HMM algorithm achieved satisfactory accuracy.	To determine the efficiency of POS tagging for named entity recognition in the Marathi language, the POS tagging system can be integrated with a named entity recognition system.
Jyoti Singh, Nisheeth Joshi, Iti Mathur [15]	The design of the POS tagger, as well as a comparison of the accuracy of the tagger's output, is exhibited in this work.	Accuracy-93.82%	The results are commendable. The system performs well, and the results obtained through their approaches are great.	Future improvements to this study would include expanding the size of the labelled corpus to increase tagging accuracy.
Jyoti Singh, Nisheeth Joshi, Iti Mathur [16]	Using HMM, developed a parts-of-speech tagging mechanism for Marathi.	Accuracy - 93.82%	For automatic tagging, the POS tagger described in this paper is quite simple and effective. The current system performs well, and the results obtained using this method are great.	Future improvements to this study would include increasing the size of the tagged corpus to improve tagging accuracy.
Vijayshri Khedkar, Pritesh Shah [17]	This work presents a Viterbi algorithm-based methodology for tagging Hindi and Marathi input.	Accuracy - 92.97% Precision - 0.93 Recall - 1.00 F-Measure - 0.9635	Considering the largest dataset in an already annotated Hindi language corpus, they have obtained good precision and accuracy.	The model is trained and evaluated on specific data (mostly news and paper articles). If the system is given data that does not match these characteristics, accuracy rates may suffer.

where w is word and t is tag of the word.

Viterbi is the decoding algorithm used in HMM. [12] paper discusses how the Viterbi algorithm is used in tagging for the Marathi language.

C. Maximum Entropy Model

Maximum Entropy Tagging relies on finding the maximum entropy model. The term "maximum entropy" refers to the highest level of unpredictability or the least amount of added structure. The tags and their probability are the results of maximum entropy tagging. The maximum entropy model defines a collection of environmental features for tag prediction. Unlike HMMs, where the ongoing observation is only influenced by the present state, a MEM's present observation can be influenced by the past state as well.

D. Conditional Random Field

CRF is a probabilistic framework for categorizing and segregating data using probabilistic approaches, such as trees and sequences. CRFs outperform generative models like the HMM for sequence labelling. Because CRFs are not constrained by the independence constraint, they can rely on random, dependent features while ignoring the distribution of such dependencies.

E. Prototyping Model

Prototypes are more meaningful and have a better assessment than clusters (due to their modest size). A few samples or prototypes are collected and then disseminated across the unseen data corpus in this model. This paradigm does not necessitate the use of a vocabulary. The log-linear model parameters are maximised via a gradient-based forward-backward search.

F. Neural Networks

A neural network is a connected group of artificial neurons using a computer model to handle pairs of input and expected responses. According to the expected responses by input characteristics, the training contexts are separated into several categories.

IV. Experimental Analysis

Fig. 4 depicts the distribution of POS tagger categories for Marathi texts among the papers examined in this study. Comparative findings for multiple models of Part of Speech Tagging Technique have been provided in Fig. 5. In the papers reviewed for this survey, different types of POS taggers were utilised to achieve POS tagging for Marathi texts. Their accuracy is listed in the Table I. Several references that use a variety of pos tagging techniques are used in this survey. Fig. 6 demonstrates the accuracy distribution of all the pos tagging approaches encountered while analyzing existing research performed to predict tags to Marathi words. The range of accuracy for each POS tagging strategy is defined by a list of accuracy values that is created by compiling

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accuracy values gathered from reference materials for each approach. As an illustration, [6] used a statistical approach and reported an accuracy of 78%. When employing the same methodology, [12] showed accuracy of 93%, whereas [14] employed statistical POS tagger to show accuracy of 86.61 percent. As a result, the surveyed statistical Marathi POS tagger has an accuracy range of 78% to 93%. From this analysis of accuracy distribution it can be understood that machine learning and deep learning-based pos taggers produce more accurate tags than conventional methods like rule-based, statistical, and hybrid POS tagging.

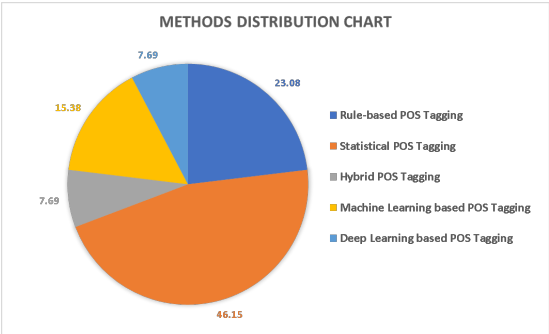

Fig. 4. Methods Distribution

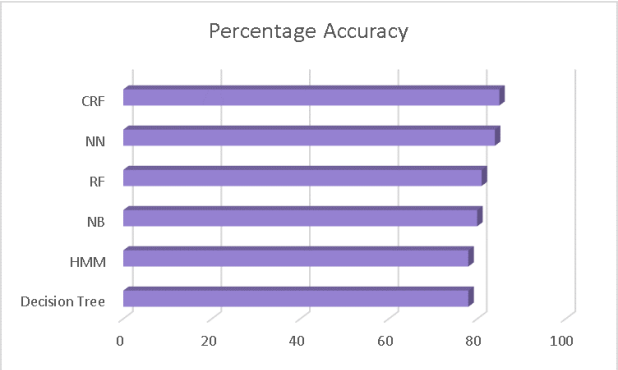

Fig. 5. Accuracy of POS taggers inspired by [6]

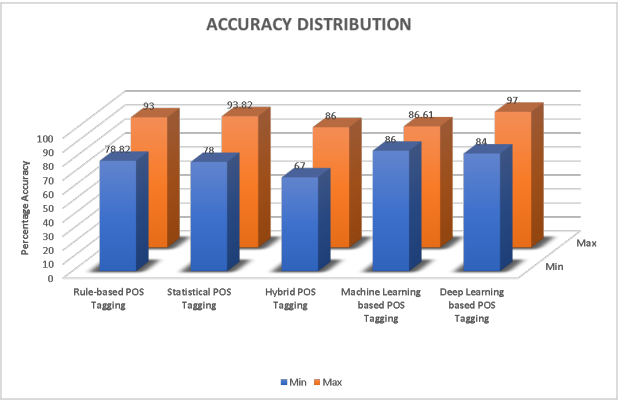

Fig. 6. Accuracy Distribution

Papers reviewed in this survey have used various performance measures to calculate the performance of proposed POS tagging algorithms. All of the assessment metrics are based on metrics used in the Confusion Matrix that provides information about the Predicted and Actual classes and includes the following metrics:

- True Positive (TP): A term that has been appropriately tagged.
- False Positive (FP): The provided term is incorrectly tagged.
- True Negative (TN): The occurrences were appropriately classified as normal occurrences.
- False Negative (FN): The provided term is not tagged to any of the tag sets.

Precision: The ratio of accurately labelled parts of speech to all tagged words in the samples:

$$Precision = \frac{TP}{TP + FP}$$

Recall: The ratio of all samples that have been accurately tagged to all samples that have been tagged

$$Recall = \frac{TP}{TP + FN}$$

Accuracy: The ratio of occurrences where a component of speech has been accurately labelled to the total number of instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

F - Measure: It is the harmonic mean of the Precision and Recall.

$$F - Measure = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

V. Applications

Part of Speech tagging is used in many different fields. It is the fundamental step while performing analysis.

A. Word Sense Disambiguation

Many words have many meanings while being in the same part of speech. The tagger was used in the development of an experimental noun homograph disambiguation algorithm. The system (known as Catch-Word) uses supervised training across a massive text corpus to satisfy lexical, orthographic, and simple syntactic facts for each sense of the ambiguous term. After a period of training, Catchword identifies new instances of the noun by comparing their context to previously noted cases and choosing the meaning with the most facts. [18] The tagger is used in two ways in this algorithm:

- Determine the target word's component of speech (filtering out the non-noun usages)
- Suggesting appropriate recommendations based on detected emotions.
- As part of the phrase recognition process, the context around the noun is analysed.

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B. Sentiment Analysis

To accomplish classification based on different combinations of POS, the data is pre-processed, which entails cleaning the information. After the data has been pre-processed, sentiment categorization is performed, which is expressed in a variety of POS combinations. The data will be retrieved after the selected pairing of POS-tagged words has been validated. The amount of positively labelled words is compared to the frequency of negatively tagged keywords after the extraction. [19]

C. Named Entity Recognition

Every NLP application starts with part-of-speech tagging. The amount of computational steps an algorithm requires to determine the POS tag for a given phrase determines the difficulty of computing the POS tag. All NLP applications will gain if the number of comparisons required for POS tagging can be minimised. The simplest operation in NLP is named entity recognition, which involves identifying the entity. POS tags are used to perform NER. [5]

D. Deep Learning and Machine Translations

The presence of ambiguity cannot be overlooked when labelling terms with diverse contextual meanings within a phrase. Deep learning and Machine learning based POS taggers are being evaluated as potential approaches for detecting words in a phrase across a paragraph. The initial query search results indicated POS tagging for a variety of applications using a variety of approaches, including AI-oriented, rule-based stochastic, and others. To identify more relevant research papers, the query term is re-defined by combining the terms deep learning and machine learning. [20]

E. Other applications

Some other domains where POS Tagging is used are Biomedical domain as mentioned in [21] and [22], and for Fake news detection as explained in [23].

VI. Conclusion

POS tagging is an important operation in NLP. The type of tagger approach utilised impacts the performance of a POS tagger in NLP. A lot of researchers have produced POS taggers. All the major methods have performed admirably. A brief overview of the existing methodologies that have been used to construct POS tagger tools is presented in this study. A comparative survey of different POS tagger systems for Marathi languages was presented. The most difficult task in each research project is to develop the most efficient POS tagger for a big training corpus that can function well in a variety of languages.

VII. Furture Scope and Open Challenges

As was previously mentioned, Marathi is a morphologically rich language that presents certain difficulties for POS tagging. When compared to English, a large tagged dataset for POS tagging in Marathi is more difficult to find. Utilizing the most recent taggers, such as, Machine Learning and Deep Learning models, can increase the tagging accuracy. Another difficulty is the existence of various definitions for a single term together with numerous functional tags that change depending on the context. While performing POS tagging of Marathi Language, it is crucial to be aware of these difficulties. This field requires more thorough research.

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